



Converting knowledge into Productivity: The role of intellectual property empowerment and digital economy in enhancing regional new quality productivity forces - Evidence from China

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ABSTRACT

In the context of China's ongoing digital transformation and the strengthening of intellectual property protection systems, cultivating new quality productive forces has become a key strategy for promoting high-quality economic development. This study investigates the synergistic effects of intellectual property empowerment and the digital economy on regional new quality productivity forces, drawing upon panel data from 31 Chinese provinces between 2012 and 2022. Utilizing an entropy weighting method, a comprehensive index system is constructed to evaluate new quality productivity forces, focusing on technological innovation, green transformation, and digital productivity. Utilizing a two-way fixed effects model, which controls for time-invariant characteristics and period-specific effects, we empirically examine the impacts of intellectual property empowerment and the digital economy on new quality productivity forces. Control variables include human capital, openness, industrialization level, urbanization rate, and economic development level. Robustness tests and instrumental variable approaches are applied to address potential endogeneity issues. Moreover, intellectual property protection and technological innovation serve as important mediating pathways in this relationship. The results also highlight regional heterogeneity, with industrial and service-oriented regions experiencing the most pronounced positive impacts, while resource-dependent areas demonstrate limited effects. These findings provide new theoretical perspectives and empirical evidence for strengthening intellectual property systems, advancing digital infrastructure, and promoting integrated regional innovation strategies. The study offers policy recommendations to support the coordinated development of intellectual property and the digital economy as dual engines for enhancing China's new quality productive forces.

1. Introduction

Currently, the global economy is undergoing a transformation, and countries are seizing the strategic opportunities of the new round of industrial revolution to promote the development of advanced productive forces. From the perspective of the world's modernization process, human history has experienced three waves of modernization: the agricultural revolution, the industrial revolution and the information revolution. These waves represent the development stages of the agricultural economy, industrial economy and digital economy respectively, and have become an important driving force for the jump in productivity (Grinin, 2022).

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Traditional productivity mainly refers to the production method from the agricultural era to the industrial era, with machines, equipment and manpower as the core production factors (Li et al., 2017). New productivity is a product of the digital economy era. Its core is driven by technological innovation, and its goal is to achieve high-quality economic development through key and disruptive technological breakthroughs. This production method is not only a comprehensive upgrade of traditional productivity, but also focuses on revolutionary breakthroughs in technology, innovative configuration of production factors, and in-depth transformation and upgrading of industries. Its connotation includes the optimal combination of workers, labor materials and labor objects. The core feature is the substantial increase in total factor productivity. In essence, it is the embodiment of advanced productivity. The key lies in innovation and high quality (Gao et al., 2024). In September 2023, when presiding over the symposium on “Promoting the Comprehensive Revitalization of Northeast China in the New Era”, Chinese President Xi Jinping proposed: Actively cultivate emerging industries such as new energy, new materials, advanced manufacturing, and electronic information, develop future industries, and accelerate the formation of new quality productive forces, enhance new momentum for development and achieve high-quality economic development (Zhang, 2024). Therefore, promoting the development of new productive forces is of great strategic significance. It is not only an inevitable requirement of the development of the times, but also the core path to achieve high-quality economic development.

Intellectual property rights are not only a legal tool to protect technological innovation achievements, but also an engine for optimal allocation of innovative resources. Data show that as of 2022, China has become the country with the largest number of patent applications in the world for ten consecutive years. The number of domestic invention patent authorizations in 2022 alone will reach 795,000, an increase of 5.3 % over the same period last year (National Bureau of Statistics of China, 2024). Under the global Patent Cooperation Treaty (PCT) framework, the number of patent applications submitted by China has accounted for 24.7 % of the global total, ranking first (Boeing et al., 2024). These figures highlight the importance of intellectual property rights in stimulating technological innovation and promoting technology commercialization and industrialization. At the same time, the development of new productive forces cannot be separated from the protection of the intellectual property rights system. Taking China as an example, in recent years, through the improvement of the intellectual property rights protection system, the contribution rate of technological innovation to economic development has increased significantly. According to the “China Statistical Yearbook”, China’s total social R&D investment accounted for 2.44 % of GDP in 2021, and a large number of R&D results have been protected through the intellectual property rights system and transformed into market value (National Bureau of Statistics of China, 2024). In addition, intellectual property rights play a particularly significant role in promoting the development of high-tech industries and improving international competitiveness (Xu, 2023). At present, the added value of high-tech industries accounts for more than 15 % of China’s GDP, becoming a new engine for economic growth (National Bureau of Statistics of China, 2024). On a global scale, intellectual property rights also serve strategic functions. In developed countries such as the United States, Germany and Japan, intellectual property rights have become an important means to seize the commanding heights of technology. For example, intellectual property-based industries (IP-Intensive Industries) in the United States contribute about 38 % of GDP and provide a large number of high-paying jobs (ÖZMACUN, 2023). In contrast, developing countries are gradually realizing a leap from technology introduction to independent innovation by strengthening the protection and utilization of intellectual property rights. Therefore, intellectual property rights are not only the core driving force for the development of new productive forces, but also determine a country’s position in future global economic competition. Studying the role of intellectual property rights in new productivity not only helps to understand the internal logic of innovation-driven economic development, but also provides an important theoretical basis for formulating more scientific and complete intellectual property rights policies. This is of great significance for achieving high-quality development, enhancing national competitiveness and promoting sustainable development of the global economy.

The digital economy represents the seamless integration of digital technologies with economic activities, functioning as a vital means to boost production efficiency and foster economic growth. More than a manifestation of technological innovation and industrial transformation, the digital economy plays a crucial role in various domains (Sun et al., 2024). It enhances the efficiency of financial institutions in supporting the real economy, accelerates industrial upgrading, spurs the growth of emerging industries, and improves governmental management capabilities and performance. Therefore, exploring how the digital economy can better empower the advancement of new productive forces holds both significant theoretical value and practical relevance. According to the 2023 Global Digital Economy White Paper released by the China Academy of Information and Communications Technology, the digital economy has emerged as a transformative force in the global industrial structure and economic landscape, driving the new wave of technological revolutions and industrial change. The report highlights that China’s digital economy achieved an impressive compound annual growth rate (CAGR) of 14.2 %, which is 1.6 times faster than the digital economy growth rate of major economies such as the United States, Germany, Japan, and South Korea (Ministry of Industry and Information Technology, 2024). This growth showcases China’s distinct advantage in digital economic development. Notably, the digital economy enhances new productive forces through its empowerment by digital technologies, driving multidimensional improvements (Wang, Lin, et al., 2024). It not only serves as a new engine for economic growth but also acts as a fundamental pillar for advancing new productive forces to higher levels.

Despite growing interest in the fields of intellectual property and the digital economy, existing research has predominantly focused on their individual impacts on specific dimensions of productivity. However, there has been limited systematic exploration of their synergistic effects in enhancing productivity. The depth and breadth of related studies remain areas for further development. This study addresses the gap by proposing an innovative dual-driver framework that identifies intellectual property empowerment and the digital economy as key drivers of new productive forces. Through both theoretical analysis and empirical testing, the research investigates how these two factors achieve synergy in improving the quality and efficiency of productivity. For the first time, this study employs the entropy method to construct a comprehensive measurement index for new productive forces (Zou et al., 2006). This index incorporates considerations of environmental constraints, resource efficiency, technological innovation, and economic benefits,

addressing the limitations of traditional productivity evaluation frameworks that often overlook quality dimensions. Moreover, this study quantitatively examines the mechanisms through which intellectual property and the digital economy influence productivity. It reveals the specific pathways and processes through which they optimize resource allocation, commercialize technological achievements, and drive industrial transformation and upgrading. This approach provides fresh perspectives and methodologies for theoretical innovation, enriching the understanding of intellectual property and the digital economy while filling a significant gap in the academic study of new productive forces.

The structure of this paper is as follows: Section 2 reviews the literature on the roles of IP empowerment and the digital economy, including their interactions in new productive forces, establishing a theoretical foundation. Section 3 outlines the research methodology and key variables. Section 4 presents the empirical analysis, detailing the results. Section 5 discusses the findings, explaining how each variable contributes to the development of new productive forces. Finally, the Section 6 summarizes the study, offers practical recommendations for fostering new productive forces, and provides future outlooks.

2. Conventional facts, Theoretical framework, hypothesis and innovation

2.1. New quality productivity forces

New quality productive forces refer to an advanced state of productivity where innovation plays a leading role, breaking away from traditional economic growth models and development paths (DENG et al., 2024; Fleeer & Li, 2023). Characterized by high technology, high efficiency, and high quality, new productive forces align with modern development principles (Jiayang, 2024). They emerge from revolutionary technological breakthroughs, innovative configurations of production factors, and profound industrial transformation and upgrading (Jianhui et al., 2024). Their essence lies in the significant enhancement of total factor productivity (TFP), with their core defined by innovation, their hallmark being superior quality, and their fundamental nature as advanced productive forces (Zheng et al., 2024). The “new” in new productive forces is fundamentally driven by technological innovation. By fostering breakthroughs in the foundational elements of productivity, technological innovation optimizes and redefines production factors. It injects powerful momentum into the development of strategic emerging industries and future-oriented industries while providing robust support for the transformation of economic structures and the advancement of industrial upgrading (Shan & Zongying, 2024; Guangqin & Mengjiao, 2024).

The advanced state of new quality productive forces emerges from the dynamic optimization and integration of three foundational elements, each playing a critical role in redefining productivity. Firstly, the workforce has shifted from traditional workers to a new generation of highly educated, multidisciplinary, and innovative talents who are able to combine knowledge expertise with digital technology. These new quality workers not only have deep knowledge reserves and professional skills, but also have creativity, adaptability, and the ability to master cutting-edge tools, which is the key to promoting high-tech and high-quality productivity (Li, 2024; Piao, 2024). Secondly, the scope of labor objects has expanded from traditional natural resources such as land, minerals, and forests to emerging and transformative fields. These fields cover digital data, clean energy, life sciences, aerospace, and deep-sea exploration, forming a hybrid pattern where physical and virtual elements coexist. This shift enables the industry to explore new areas, optimize resources, and develop innovative solutions that meet sustainable development goals (Battistoni, 2017; Singh, 2013). Thirdly, labor tools have undergone profound evolution, transforming from traditional mechanical and manual tools to intelligent, environmentally friendly systems driven by advanced algorithms and computing power. These new productivity tools focus on intelligence, efficiency, precision, and environmental sustainability, transforming production processes into intelligent, multifunctional operations to ensure higher safety standards and resource conservation (ElMaraghy et al., 2021). By seamlessly integrating these three dimensions - innovative workers, advanced labor objects and intelligent labor tools, the new productivity embodies a transformative transformation, promotes technological breakthroughs, optimizes resource utilization, and accelerates the transition to a high-tech, sustainable economic model. This holistic framework not only drives productivity growth, but also lays the foundation for future economic resilience and innovation (Mindell & Reynolds, 2023; Kumar et al., 2024).

2.2. Theoretical examination and the establishment of hypotheses concerning the application of intellectual property empowerment and digital economy in the context of new quality productivity forces development

Intellectual property empowerment refers to the process of achieving optimal resource allocation and value appreciation through the creation, protection, application and transaction of intellectual property (Chen, Yin, et al., 2024). Existing research shows that intellectual property rights are a core component of the innovation system, which not only provides companies with market competitive advantages but also promotes the effective flow of knowledge through patent transactions and technology transfer (Duan et al., 2023). In addition, there is a significant positive relationship between the intensity of intellectual property protection and innovation capabilities (Wiroonrath et al., 2024), indicating that a high level of intellectual property empowerment may enhance the innovation level of an economy, thereby promoting productivity growth. New quality productivity forces is different from traditional productivity indicators. It pays more attention to the upgrading and optimization of production structure through technological innovation, organizational innovation and institutional innovation (Abbas et al., 2023). In the process of studying productivity changes, scholars have pointed out that the key factors affecting new productivity include technological innovation input (Ji et al., 2023; Awan & Yaqoob, 2023), knowledge spillover effects (Wang & Choi, 2023), and policy support environment (Jin et al., 2023). Institutional economics points out that a good institutional environment can improve the efficiency of resource allocation, and the intellectual property system is an important part of the innovative economic system (Pisuke & Kelli, 2008). In the process of

intellectual property empowerment, tools such as patents and copyrights can not only promote knowledge sharing, but also enhance collaborative innovation capabilities among industries through spillover effects (Lemley, 1996).

Although many studies have explored the relationship between intellectual property and innovation, few literatures have systematically analyzed how intellectual property empowerment affects new quality productivity forces but lacks in-depth analysis of the multi-dimensional impact mechanism of intellectual property empowerment. At the same time, empirical research on how intellectual property empowerment drives new productivity in the Chinese context is even scarcer, which provides an important research entry point for this study. Therefore, we made the hypothesis.

H1a. Intellectual Property Empowerment (IPEn) has a significant positive impact on New Quality Productivity Forces (NQPF)

The digital economy has become a force for change, significantly improving productivity and cultivating new high-quality productivity in various industries. This shift is particularly evident in the context of green total factor productivity (GTFP), where the digital economy not only improves economic efficiency but also promotes sustainable practices. Research shows that the digital economy promotes green innovation, which is crucial to increasing GTFP. For example, research shows that cities in the Yangtze River Delta and other regions have achieved significant green economic growth through the development of the digital economy, highlighting the role of digital technology in driving green innovation (Wang et al., 2024; Sheng, Zhu, & Chen, 2024). In addition, the integration of digital technology and traditional industries has brought about significant improvements in productivity. The digital economy is characterized by its reliance on digital knowledge and information as key production factors. This integration extends the industrial chain and gives birth to new industries, thereby upgrading existing industries (Liu et al., 2022; Liu, Yang, Li, & Zhong, 2022a, 2022b; Zhao et al., 2023). Empirical evidence shows that digital infrastructure contributes significantly to total factor productivity, with positive spatial spillover effects observed in different regions (Ma et al., 2023; Zhao et al., 2022).

At present, a large amount of research focuses on the impact of the digital economy on traditional economic growth (Zhang et al., 2021), but the micro-mechanical analysis of how the digital economy improves new productivity is still insufficient, especially in the systematic development of digital infrastructure. There are still deficiencies in empirical research. At the same time, research on the impact mechanism of new productivity in the context of China's digital economy is also relatively scarce. Therefore, we set up hypothesis.

H1b. Digital economy (DigEcon) has a significant positive impact on New Quality Productivity Forces (NQPF)

Intellectual property protection plays a key role in improving new quality productivity forces. Existing research shows that strengthened intellectual property protection can promote technological innovation and further improve production efficiency and quality (Alikhan & Mashelkar, 2009). Specifically, intellectual property protection can encourage companies to increase investment in R&D, thereby accelerating the commercialization of technological innovation results (Park et al., 2024). In addition, by building an intellectual property market, technology transfer and cooperative innovation among enterprises can be strengthened, forming resource integration and spillover effects, and ultimately promoting the overall improvement of new productivity.

Currently, research on how intellectual property empowerment promotes new quality productivity forces through intellectual property protection is still a major gap in academic research. Existing literature mainly focuses on how intellectual property protection stimulates innovation (Qiu & Yu, 2010; Chen & Wu, 2022), promotes technology diffusion and economic efficiency (Chen & Chen, 2024), and strengthens the impact of intellectual property protection on corporate R&D investment, technology output efficiency and enhancing effect (Deng et al., 2019; Kumar, 1996). At the same time, some studies have revealed the important role that intellectual property protection plays in technology transfer, international patent cooperation and the digital economy, especially its long-term impact on high-tech industries in the context of globalization. However, the existing literature has not yet conducted a systematic analysis on how intellectual property empowerment, as a knowledge-driven strategy, promotes the formation and optimization of new quality productivity forces with technological innovation, high added value and sustainable development as its core through the multi-level application of intellectual property protection. In order to fill the research gap in this field, this article puts forward a hypothesis based on the theoretical foundation and practical application scenarios of intellectual property empowerment.

H2a. Intellectual Property Empowerment (IPEn) can promote New Quality Productivity Forces (NQPF) through Intellectual Property Protection (IPP)

Research shows that digital infrastructure significantly increases total factor productivity (TFP), a key indicator of economic efficiency (Pan et al., 2022; Tang & Zhao, 2023). The integration of digital technologies not only improves operational efficiency, but also reduces transaction costs, promotes smoother market interactions, and promotes higher productivity levels across departments (Rehman & Nunziante, 2023; Li et al., 2024). As businesses leverage digital tools, protecting their innovations through intellectual property rights becomes increasingly important to maintain a competitive advantage and secure market position. However, it is important to recognize the dual nature of intellectual property protection. Although strong intellectual property rights can stimulate innovation and economic growth, over-protection may lead to monopolistic behavior, thereby stifling competition and hindering technological progress ("From the perspective of intellectual property rights protection, FDI in producer services and the upgrading of the Pearl River Delta manufacturing industry") (Sun, Zhang, & Zhu, 2023). This paradox requires a balanced approach to intellectual property rights policy, ensuring that protection is sufficient to encourage innovation without creating barriers to market entry for new and existing companies.

Furthermore, the impact of intellectual property rights protection extends beyond the domestic market. Countries with robust intellectual property rights regimes are more likely to attract foreign direct investment (FDI) as investors seek to ensure that their technologies and innovations are protected (Adams, 2010). The influx of FDI can further improve local productivity through the

introduction of advanced technologies and practices, thus promoting the overall economic development of the host country (Dai, 2020; Sun, Zhang, & Zhu, 2023). Therefore, we assume.

H2b. Digital economy (DigEcon) can promote new quality productivity forces (NQPF) through Intellectual Property Protection (IPP)

The relationship between intellectual property (IP) rights and innovation is well-documented, with empirical evidence suggesting that stronger IP protections can lead to increased investment in research and development (R&D) and subsequently foster innovation across various sectors (Ayerbe et al., 2024; Song et al., 2024). Bielig (2012) highlights that the impact of IP on economic development varies by sector, but overall, it suggests a positive influence on innovation generation. This is echoed by Li (2024), who emphasizes that comprehensive intellectual property capabilities—encompassing creation, application, protection, and management—are essential for firms to enhance their productivity and competitive strength.

Moreover, the role of IP in facilitating FDI is significant. Dussaux et al. (2022) argue that strengthening IP rights can help developing countries attract foreign capital and advanced technologies, thereby enhancing their industrial capabilities. This relationship underscores the importance of a balanced approach to IP protection, as excessive restrictions can stifle innovation and economic growth (Luo & Zhao, 2024). However, current research mainly focuses on the single action mechanism of intellectual property on technological innovation or economic growth, and there is relatively insufficient analysis of the path of how intellectual property empowerment systematically promotes new quality productivity forces through technological innovation. To fill this research gap, this paper hypothesizes that.

H3a. Intellectual Property Empowerment (IPEn) can promote New Quality Productivity Forces (NQPF) through Technological innovation (TI)

One of the primary mechanisms through which the digital economy enhances productivity is through technological innovation. Digital technologies facilitate the development of new business models and improve existing processes, thereby driving efficiency. Existing literature shows that the digital economy can effectively promote the optimization of economic structure and the development of high value-added industries through technological innovations such as big data, artificial intelligence, and the Internet of Things (Teece, 2018; Ding et al., 2021; Lyu et al., 2023). For instance, the introduction of digital infrastructure and services, such as cloud computing and industrial internet applications, has been linked to improved performance in traditional industries (Hao et al., 2023; Li et al., 2022). The nonlinear effects of the digital economy on productivity, as highlighted in recent studies, suggest that there are thresholds beyond which the benefits of digitalization become more pronounced (Guo et al., 2022; Wang, Lin, et al., 2024; Zhang et al., 2022). For example, regions that have invested significantly in digital infrastructure tend to experience greater improvements in productivity compared to those that have not reached similar levels of investment (Andrews et al., 2018; Chen, Yin, et al., 2024). This threshold effect emphasizes the importance of creating conducive environments for digital innovation, which can lead to significant economic benefits. Thus, we hypothesize below.

H3b. Digital economy (DigEcon) can promote New Quality Productivity Forces (NQPF) through Technological innovation (TI)

2.3. Innovation and entry point

This study focuses on all provinces in China as the research context, employing innovative measurement indicators and empirical methods to uncover the distinct pathways through which intellectual property empowerment and the digital economy influence new quality productive forces in the Chinese setting. The primary objective is to construct an innovative analytical framework and provide an in-depth examination of how intellectual property empowerment and the digital economy collaboratively enhance new productive forces. Specifically, the research identifies three key directions: first, investigating the direct effects of intellectual property empowerment and the digital economy on new quality productive forces; second, analyzing the mediating roles of intellectual property protection and technological innovation in this relationship; and third, examining heterogeneity, with a focus on regional disparities in the impacts of intellectual property empowerment and the digital economy on new quality productivity forces development.

In existing research, intellectual property empowerment and digital economy are the core factors of new quality productivity forces, and their single mechanism of action has been widely discussed. However, the discussion on how the two can exert a synergistic effect in improving productivity is still relatively limited. At the same time, the current literature mostly focuses on the single impact of intellectual property protection on technological innovation or economic growth (Ciocoiu, 2011; Kim et al., 2012), and the promotion effect of the digital economy on traditional productivity improvements (Liu et al., 2022; Liu et al., 2022a, 2022b; Su et al., 2021). There is still insufficient research on the mechanism of promoting new quality productivity forces in a multi-level manner, which further provides research space for this study. Unlike existing literature, this study is the first to systematically reveal the synergistic pathways through which intellectual property empowerment and the digital economy enhance new quality productivity forces, while incorporating regional heterogeneity analysis to fill the gap in the Chinese context.

3. Methodology

3.1. Data source and sample

The sample data of our paper comes from the China Statistical Yearbook and the provincial statistical yearbooks (National Bureau of Statistics of China, 2024), covering 31 provinces (including municipalities and autonomous regions) from 2012 to 2022, with a total

sample size of 341 provincial-year data, forming a balanced panel data set. The reason for selecting this time span is that China's digital economy has developed rapidly and the intellectual property protection policy has been continuously improved during this period, which can better reflect the impact of digital economy and intellectual property empowerment on new quality productivity. In order to alleviate the potential impact of outliers in the data on the regression results, this paper shrinks the data by 1 % above and below. This method can effectively reduce the interference of extreme values on the analysis results and ensure the robustness and reliability of the regression analysis.

3.2. Variable selection and definition

3.2.1. Explained variables

As the explained variable of this study, the new quality productivity forces (NQPF) aims to measure the comprehensive productivity level of regions or enterprises in terms of innovation-driven and technological progress under the background of digital economy and intellectual property empowerment (Zhang et al., 2024). The new quality productivity not only reflects the traditional production efficiency, but also includes the contribution of emerging factors such as technological innovation, green transformation, and industrial upgrading to productivity improvement. In order to measure the new quality productivity forces scientifically and objectively, this paper adopts the entropy method for multi-indicator synthesis. The entropy method can effectively reduce the influence of subjective factors and ensure the objectivity of the comprehensive evaluation by assigning weights to each indicator (Ding et al., 2016). Specifically, the composition of the new quality productivity forces includes multiple dimensions such as innovation capability, technical efficiency, and resource utilization efficiency. The entropy method is used to integrate the information of each indicator to form a comprehensive score, which can more comprehensively reflect the overall level of the new quality productivity forces (Table 1).

3.2.2. Mechanism variables

We consider technological innovation (TI) and intellectual property protection (IPP) as our mechanism variables. Technological innovation is one of the key factors in promoting the improvement of new quality productivity forces. It refers to achieving

Table 1

New quality productivity forces indicator system.

Primary Category	Secondary Categories	Indices	Tertiary Category	Definitions	Units	Attributes
Technological Productivity	Innovative Production Capacity	A1	Innovative Development	Number of domestic patents granted	Units	Positive
		A2	Innovative Industry	Revenue from high-tech industry	Ten thousand yuan	Positive
		A3	Innovative Products	Innovative product output value of regulated industrial enterprises	Ten thousand yuan	Positive
	Technical Production Efficiency	A4	Technical Efficiency	Labor productivity of regulated industrial enterprises	%	Positive
		A5	Technical Research and Development (R&D)	Full-time R&D personnel in regulated industrial enterprises	Person-hours	Positive
		A6	Technical Productivity	Robot installation density	%	%
Green Productivity	Resource-Saving Production Capacity	B1	Energy Consumption Intensity	Energy consumption per unit of GDP	%	Negative
		B2	Energy Structure	Share of chemical energy in total energy consumption	%	Negative
		B3	Water Usage Intensity	Industrial water use per unit of GDP	%	Negative
	Environmental Friendliness	B4	Solid Waste Utilization	Reutilization rate of industrial solid waste	%	Positive
		B5	Wastewater Emissions	Industrial wastewater discharge per unit of GDP	%	Negative
		B6	Waste Gas Emissions	Industrial SO ₂ discharge per unit of GDP	%	Negative
Digital Productivity	Digital Industrial Capacity	C1	Electronic Information Manufacturing	Revenue from clustered electronic information industries	Ten thousand yuan	Positive
		C2	Telecommunications Business Information	Telecommunications business revenue	Ten thousand yuan	Positive
	Industrial Digitalization	C3	Internet Development	Broadband penetration rate at the client end	%	Positive
		C4	Software Services	Revenue from software business	Ten thousand yuan	Positive
		C5	Digital Information	Fiber-optic cable length	Meter	Positive
		C6	E-Commerce Services	E-commerce sales revenue	Ten thousand yuan	Positive

breakthroughs in products, processes, management, etc. through technological research and development, technological transformation and innovative practices, thereby promoting the upgrading of productivity and improving efficiency (Shen & Li, 2024). In this study, TI is measured by the innovation index in the report. The innovation index is a multi-dimensional indicator that evaluates the level of technological innovation by comprehensively considering the input, output and conversion efficiency of a region or enterprise in technological innovation (Peng et al., 2021). Specifically, the innovation index comprehensively considers the following aspects: R&D investment (such as R&D funds, number of R&D personnel), technical patents (such as the number of patent applications, number of patent authorizations), technical product output (such as the output value of high-tech products), and technology conversion capabilities (such as the degree of industrialization of technological achievements).

Intellectual property protection (IPP) is one of the core mechanisms to promote technological innovation and economic transformation. Strong intellectual property protection can provide legal protection for innovation, encourage enterprises and individuals to actively invest in technological research and development and innovation activities, and thus improve overall productivity (Yi et al., 2024). IPP is not only a legal constraint, but also directly affects the speed and breadth of transformation of innovative achievements. An effective protection system can reduce the risk of technology leakage and abuse, enable innovators to obtain sufficient returns from their research and development activities, and thus enhance their motivation for continuous innovation. The intellectual property protection index in this study is used to measure the strength of a region or country in intellectual property protection (Hou et al., 2023).

3.2.3. Explanation variables

The explanation variables of our study are intellectual property empowerment (IPEn) and digital economy (DigEcon). Intellectual property empowerment refers to the process of promoting technological innovation, industrial upgrading and economic development through the effective use of intellectual property (Disemadi, 2022). In this study, IPEn is measured by successful patent transfer cases, specifically the ratio of the number of successfully transferred patents to the total number of patents. This indicator can reflect the ability of a region or enterprise to use intellectual property for technological innovation and marketization (Li & Chen, 2024; Ma et al., 2024).

Digital economy (DigEcon) is a new economic form that promotes economic development based on digital technology, with the help of the Internet, information technology and digital transformation (Kerras et al., 2020). It plays an important role in promoting industrial upgrading, improving production efficiency, promoting innovation and green development (Kilimova et al., 2021). DigEcon in this study is measured by the digital economic development index (Kerras et al., 2020). The index system includes three aspects: informatization development, Internet development and digital transaction development. The index comprehensively considers indicators of informatization infrastructure construction such as optical cable density, mobile phone base station density, and the proportion of informatization practitioners, indicators of informatization impact such as total telecommunications business volume and software business revenue, and indicators of Internet development such as fixed and mobile Internet infrastructure and Internet user penetration rate. At the same time, digital transaction foundations and impact indicators such as the proportion of corporate websites and e-commerce sales are also considered. All these indicators are positive. The Digital Economic Development Index reflects the level of digital economy in a given region, with higher index values indicating more advanced development. This index is derived by applying the entropy method to weight and synthesize various indicators. The complete indicator system is detailed in Table 2.

Table 2
Digital economic indicator system.

Main Indicator	First-level Indicator	Second-level Indicator	Measurement Indicator	Indicator Attribute
Digital Economy Development Index	Informatization Development Indicators	Information technology foundation	Fiber Density	Positive
			Mobile Telephone Base Station Density	Positive
			Proportion of IT Industry Employees	Positive
		Informatization impact	Total Telecom Business Volume	Positive
			Software Business Revenue	Positive
			Fixed Internet Access Terminal Density	Positive
	Internet Development Indicators	Fixed-end Internet infrastructure	Mobile Telephone Penetration Rate	Positive
		Mobile Internet Basics	Proportion of Broadband Internet Users	Positive
		Fixed-end Internet impact	Proportion of Mobile Internet Users	Positive
		Internet Development Impact	Proportion of Enterprise Websites	Positive
	Digital Transaction Development Index	Digital Trade Foundation	Proportion of Enterprises Using Computers	Positive
			Proportion of E-commerce Utilization	Positive
			E-commerce Sales Revenue	Positive
		Digital Trade Impact	Online Retail Sales Amount	Positive

3.2.4. Control variables

In this study, several control variables were selected to control other factors that may affect new quality productivity (see Table 3). These variables include: Human capital level (HCL), which is measured by the number of college students per 100,000 people, reflecting the educational level of regional human resources (Nyberg et al., 2014). Level of Openness (Open) is measured by the ratio of total import and export volume to GDP. This indicator reflects the degree of a region's economic dependence on foreign trade (Huchet-Bourdon et al., 2018). Industrialization level (Indus) is measured by the proportion of industrial added value to regional GDP, reflecting the degree of regional industrialization (Wang et al., 2018). Urbanization level (Urban) is measured by the proportion of urban population to total population, representing the urbanization process of a region (Davis, 2015). Finally, Level of Economic Development (LnGDP) is measured by the logarithmic value of per capita GDP, which reflects the economic development level of a region (Duan et al., 2021).

3.3. Model design

In this study, a two-way fixed effects model is applied to empirically investigate the influence of the digital economy and intellectual property empowerment on the quality of new productivity forces (De & d'Haultfoeuille, 2020). This model effectively accounts for both individual and time-specific effects, reducing the risk of omitted variable bias and thereby improving the reliability of the regression results.

3.3.1. Entropy method

The entropy method is an objective weighting technique widely applied in comprehensive evaluations involving multiple indicators. Rooted in information theory, the method is based on the concept of entropy, which measures the degree of disorder or uncertainty associated with an indicator. In this context, a lower entropy value indicates greater variability and more useful information provided by the indicator, thus deserving a higher weight in the evaluation system (Jaynes, 1982). Specifically, the entropy method involves several key steps:

$$e_j = -k \sum_{i=1}^n p_{ij} \ln(p_{ij}) \quad (1)$$

where $k = \frac{1}{\ln(n)}$

Here, p_{ij} represents the normalized value of the i -th observation under the j -th indicator, and n is the total number of observations. In cases where $p_{ij} = 0$, the term $p_{ij} \ln(p_{ij})$ is conventionally set to zero to avoid undefined results. The degree of divergence, also known as entropy redundancy, is subsequently computed as:

$$d_j = 1 - e_j \quad (2)$$

Finally, the objective weight of indicator j , denoted as w_j , is obtained by normalizing the divergence values:

$$w_j = \frac{d_j}{\sum_{j=1}^m d_j} \quad (3)$$

3.3.2. Benchmark regression model

In order to study the impact of digital economy and intellectual property empowerment on new quality productivity, this study first constructed a baseline regression model. This model aims to examine how the two variables of digital economy and intellectual property empowerment directly affect new quality productivity forces (NQPF) (Hong, Wang, & Willis, 2011, July) and the basic form of the benchmark regression model are:

$$NQPF_{it} = \alpha_0 + \alpha_1 IPEn_{it} + \alpha_2 X_{it} + \mu_i + \nu_t + \varepsilon_{it} \quad (4)$$

Table 3
Descriptive statistics.

VarName	Obs	Mean	SD	Min	Median	Max
NQPF	341	0.1990	0.181	0.03	0.1465	0.8768
TI	341	28.6939	10.708	15.78	26.0500	65.4900
IPP	341	0.6331	0.137	0.43	0.6215	0.9864
IPEn	341	0.0308	0.062	0.00	0.0183	0.5621
DigEcon	341	0.1244	0.100	0.02	0.0918	0.5903
HCL	341	0.0209	0.006	0.01	0.0206	0.0436
Open	341	0.2593	0.266	0.01	0.1432	1.3541
Indus	341	0.3197	0.088	0.07	0.3271	0.5418
Urban	341	0.5981	0.127	0.23	0.5900	0.9000
LnGDP	341	10.8966	0.445	9.85	10.8575	12.1547

$$\text{DigEcon}_{it} = \alpha_0 + \alpha_1 \text{IPEn}_{it} + \alpha_2 X_{it} + \mu_i + \nu_t + \varepsilon_{it} \quad (5)$$

Among them, NQPF_{it} represents the new quality productivity forces of the i province in the t year, as the explained variable; DigEcon_{it} is the digital economy development index, which measures the overall development level of the region in the field of digital economy; IPEn_{it} is intellectual property empowerment, which represents the degree of market-oriented application of intellectual property, specifically expressed as the ratio of the number of successfully transferred patents to the total number of patents; α_0 is a constant term; α_1 and α_2 are the regression coefficients of digital economy and intellectual property empowerment, respectively, indicating their impact on new quality productivity. μ_i is an individual fixed effect, controlling for time-invariant characteristic differences between different regions; ν_t is a time fixed effect, controlling for external shocks experienced by all individuals in the same time period, such as macroeconomic changes or policy adjustments; ε_{it} is a random error term. The core purpose of the benchmark regression model is to analyze how the digital economy and intellectual property empowerment directly affect the improvement of new quality productivity forces.

3.3.3. Mechanism model

Building on the theoretical analysis outlined earlier, the digital economy (DigEcon) and intellectual property empowerment (IPEn) are hypothesized to influence the new quality of productivity forces (NQPF) through distinct pathways. Specifically, DigEcon and IPEn may indirectly enhance NQPF via two key mediating variables: technological innovation (TI) and intellectual property protection (IPP). To examine these potential mechanisms, this study develops a mechanism model grounded in the Benchmark regression framework. Drawing on the methodological approach of Vareda (2023), the mechanism model is constructed to empirically test whether DigEcon and IPEn affect NQPF through these mediating factors.

$$\text{TI}_{it} = \gamma_0 + \gamma_1 \text{IPEn}_{it} + \gamma_2 X_{it} + \gamma_i + \nu_t + \varepsilon_{it} \quad (6)$$

$$\text{IPP}_{it} = \eta_0 + \eta_1 \text{IPEn}_{it} + \eta_2 X_{it} + \mu_i + \nu_t + \varepsilon_{it} \quad (7)$$

$$\text{TI}_{it} = \kappa_0 + \kappa_1 \text{DigEcon}_{it} + \kappa_2 X_{it} + \gamma_i + \nu_t + \varepsilon_{it} \quad (8)$$

$$\text{IPP}_{it} = \lambda_0 + \lambda_1 \text{DigEcon}_{it} + \lambda_2 X_{it} + \mu_i + \nu_t + \varepsilon_{it} \quad (9)$$

Step one involves using a baseline regression model to examine the direct impact of intellectual property empowerment (IPEn) and the digital economy (DigEcon) on new quality productivity forces (NQPF). This stage focuses on determining whether the digital economy and intellectual property empowerment have a significant positive effect on enhancing new quality productivity, thereby validating their direct contribution.

Step two explores the influence of intellectual property empowerment and the digital economy on intellectual property protection (IPP) and technological innovation (TI). At this stage, regression analysis is conducted to investigate how the digital economy and intellectual property empowerment drive these mediating variables, highlighting their potential roles in fostering intellectual property protection and technological innovation.

By conducting the above two-step analysis, this study employs a mechanism model to reveal how the digital economy and intellectual property empowerment indirectly promote the improvement of new quality productivity through mediating factors such as technological innovation and intellectual property protection. The findings provide both theoretical insights and empirical evidence to support policy formulation aimed at sustainable development in new quality productivity forces.

4. Analysis of empirical results

4.1. Descriptive statistics

To gain a deeper understanding of the variables in this study, descriptive statistical analysis was conducted on the sample data, providing insights into the distribution and regional differences of key factors. The mean of NQPF is 0.1990, with a standard deviation of 0.181, indicating significant disparities among provinces, as most regions exhibit low productivity levels while a few show considerably higher levels. TI has a mean of 28.6939 and a standard deviation of 10.708, reflecting substantial differences in innovation input and output across regions. Similarly, IPP shows relatively concentrated levels, with a mean of 0.6331 and most regions demonstrating strong protection. However, IPEn , with a mean of 0.0308 and a large standard deviation, highlights limited market application and notable regional variation. The DigEcon also reveals considerable disparities, with a mean of 0.1244, as some regions significantly lag in development. In contrast, HCL demonstrates minimal variation, with a mean of 0.0209. The Open , with a mean of 0.2593, shows wide-ranging regional differences, with some provinces being highly open to international trade. Similarly, Indus (mean: 0.3197) and Urban (mean: 0.5981) highlight distinct regional disparities, with urbanization differences being relatively smaller. Lastly, the LnGDP , with a mean of 10.8966, reflects a certain degree of variation in economic development, though overall levels remain relatively high. These findings underscore the substantial regional heterogeneity in key variables, providing a basis for further investigation into their impacts on the NQPF .

4.2. Correlation analysis

In order to further understand the relationship between the main variables and evaluate possible collinearity problems, this article conducted a correlation analysis on each variable in the sample (Gogtay & Thatte, 2017). The results of the correlation analysis show that there is a certain positive correlation between the digital economy, intellectual property empowerment, technological innovation and new productivity. Specifically, the correlation coefficient between the DigEcon and NQPF is 0.45, and is significant at the 1 % significance level, which shows that the development of the digital economy has a strong positive impact on the improvement of new quality productivity forces. The correlation coefficient between IPEn and NQPF is 0.35, which is also significant at the 5 % significance level, indicating that intellectual property empowerment plays a positive role in promoting new quality productivity. The correlation coefficient between TI and NQPF is 0.40, which is also significant at the 1 % significance level, indicating that technological innovation has a positive role in promoting new quality productivity forces (Table 4).

4.3. Benchmark regression

We use a Benchmark regression model to examine the direct impact of IPEn and the DigEcon on NQPF (Yang & Loog, 2018). According to hypotheses 1a and 1b, the digital economy and intellectual property empowerment are expected to positively influence new quality productivity. The primary objective of the Benchmark regression model is to validate these hypotheses. In regression model (1), the coefficient of IPEn is 0.1953 and is statistically significant at the 1 % level, indicating a significant positive impact of IPEn on NQPF. This supports hypothesis 1a, demonstrating that intellectual property empowerment contributes to the enhancement of new quality productivity forces. Similarly, in regression model (2), the coefficient of the DigEcon is 0.2827 and also significant at the 1 % level, confirming hypothesis 1b and suggesting that advancements in the digital economy facilitate the growth of new quality productivity forces. The positive and significant coefficients for both variables indicate their independent and substantial contributions to improving new quality productivity. In summary, the results of the Benchmark regression model validate the positive effects of both the digital economy and intellectual property empowerment on new quality productivity forces (Table 5).

4.4. Robustness test

4.4.1. Excluding municipalities

To test the robustness of the model results, the sample was adjusted by excluding China's four municipalities (Beijing, Shanghai, Tianjin, and Chongqing). These municipalities differ significantly from other provinces in terms of economic development, policies, and industrial structure, which could potentially influence the regression outcomes. Given their prominent roles in the national economy and their distinct advantages in digital economy development and intellectual property protection, excluding these regions helps verify whether the model results are affected by these unique areas (Mao et al., 2022). After removing the municipalities, a new regression analysis was conducted to eliminate potential biases stemming from their economic characteristics, policy implementations, and data deviations. The updated regression results remain consistent with the original findings, confirming the significant positive effects of the DigEcon and IPEn. This robustness check further supports the validity of the model's conclusions (Table 6).

4.4.2. Lagged variable analysis

To further validate the causal relationship in the model, this study applies a lagged variable approach by using the first-lagged terms of the core explanatory variables, IPEn and the DigEcon, for regression analysis (Blackwell & Glynn, 2018). The results show that the coefficient for lagged IPEn is 0.1386, while the coefficient for lagged DigEcon is 0.2173, both of which are statistically significant at the 1 % level. These findings indicate that the positive effects of IPEn and the DigEcon persist over time, further supporting hypotheses 1a and 1b. Moreover, the direction and significance of the control variables remain consistent with those in the Benchmark regression, underscoring the robustness of the model specification (Table 7).

Table 4
Correlation analysis.

	NQPF	TI	IPP	IPEn	DigEcon	HCL	Open	Indus	Urban	LnGDP
NQPF	1									
TI	0.907***	1								
IPP	−0.602***	−0.656***	1							
IPEn	0.470***	0.418***	−0.225***	1						
DigEcon	0.857***	0.861***	−0.595***	0.526***	1					
HCL	0.0710	0.214***	−0.336***	0.0400	0.222***	1				
Open	0.679***	0.800***	−0.498***	0.211***	0.648***	0.251***	1			
Indus	0.230***	0.120**	−0.307***	0.0770	0.0310	0.0670	−0.0220	1		
Urban	0.506***	0.641***	−0.487***	0.188***	0.615***	0.608***	0.747***	0.0610	1	
LnGDP	0.580***	0.683***	−0.613***	0.229***	0.745***	0.509***	0.648***	−0.0710	0.849***	1

Table 5
Benchmark regression model.

VARIABLES	(1)	(2)
	NQPF	NQPF
IPEn	0.1953*** (4.6854)	
DigEcon		0.2827*** (4.7154)
HCL	−0.5786 (−0.3655)	1.2900 (0.7891)
Open	−0.1288*** (−3.2086)	−0.1338*** (−3.3664)
Indus	−0.1299 (−1.3761)	−0.0438 (−0.4440)
Urban	0.1418 (0.8434)	0.3175* (1.8792)
LnGDP	0.1528*** (3.5992)	0.1026** (2.3809)
Constant	−1.4147*** (−3.4661)	−1.0630*** (−2.5998)
Observations	341	341
R-squared	0.283	0.284
Number of code	31	31
Pro	YES	YES
Year	YES	YES

t-statistics in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

Table 6
Robustness test excluding municipalities.

VARIABLES	(1)	(2)
	NQPF	NQPF
IPEn	0.1728*** (3.6135)	
DigEcon		0.2051*** (3.3207)
HCL	−0.1673 (−0.0934)	−0.5586 (−0.4060)
Open	−0.1875*** (−3.3008)	−0.2038*** (−4.2250)
Indus	−0.1291 (−1.2379)	0.0116 (0.1348)
Urban	0.1759 (0.7142)	−0.0426 (−0.1964)
LnGDP	0.1390*** (2.8982)	0.0126 (0.3414)
Constant	−1.2808*** (−2.9220)	0.0979 (0.3532)
Observations	297	297
R-squared	0.260	0.204
Number of code	27	27
Pro	YES	YES
Year	YES	YES

t-statistics in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

Table 7
Lagged processing for robustness test.

VARIABLES	(1)	(2)
	F.NQPF	F.NQPF
IPEn	0.1386*** (3.4837)	
DigEcon		0.2173*** (3.5185)
HCL	−0.6187 (−0.3668)	0.8435 (0.4851)

(continued on next page)

Table 7 (continued)

VARIABLES	(1)	(2)
	F.NQPF	F.NQPF
Open	−0.1127*** (−2.8667)	−0.1180*** (−3.0418)
Indus	0.1072 (1.0571)	0.1561 (1.4983)
Urban	0.2432 (1.4466)	0.3821** (2.2533)
LnGDP	0.1307*** (3.1248)	0.0921** (2.1664)
Constant	−1.3175*** (−3.2819)	−1.0440** (−2.5927)
Observations	310	310
R-squared	0.233	0.234
Number of code	31	31
Pro	YES	YES
Year	YES	YES

t-statistics in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

4.5. Endogeneity test

4.5.1. Propensity score matching (PSM)

To address potential endogeneity issues and reduce endogeneity bias, this study employs the 1:1 nearest-neighbor matching method using propensity score matching (PSM). Endogeneity problems often arise due to omitted variables, sample selection bias, or bidirectional causality, potentially leading to inconsistent regression results. To resolve this issue, PSM is applied to match individuals with similar characteristics, thereby eliminating sample selection bias and making the treatment and control groups more comparable, ultimately enhancing the reliability of regression results. The implementation of PSM involves two steps. First, the propensity scores (i. e., the probability of receiving treatment) for each observation are calculated. Second, 1:1 nearest-neighbor matching is conducted, pairing each treated sample with the most similar untreated sample. After matching, only the matched samples are used for regression analysis to ensure that the relationship between the DigEcon, IPEn, and NQPF is no longer affected by endogeneity bias. The regression results based on the matched sample align with the baseline findings. In model (1), the coefficient for IPEn is 0.1793 and statistically significant at the 1 % level, indicating a robust positive impact of IPEn on NQPF. In model (2), the coefficient for the DigEcon is 0.1331, significant at the 5 % level, demonstrating that the digital economy also has a positive influence on new quality productivity forces. These results further confirm the reliability of the causal relationships identified in this study (Table 8).

4.5.2. Instrumental variables two-stage least squares (IV-2SLS)

To address endogeneity issues, this study employs the two-stage least squares (2SLS) regression analysis. Endogeneity problems often arise due to bidirectional causality, omitted variables, or sample selection bias, which can lead to inconsistent regression results (Mann, 2006). To resolve these issues, two instrumental variables (IVs) were selected: the number of patent applications in each province as an instrument for intellectual property empowerment (IPEn) and the distance to Hangzhou as an instrument for the digital economy (DigEcon). The selection of these instrumental variables is well-justified. The number of patent applications is closely related to intellectual property empowerment but does not directly influence NQPF, thus effectively addressing endogeneity bias. Similarly, the distance to Hangzhou is highly correlated with the development of the digital economy, given Hangzhou's central role in China's digital economy, but has no direct impact on new quality productivity, making it a valid exogenous variable. By using 2SLS, this study ensures that the causal relationships between IPEn, DigEcon, and NQPF are accurately estimated, thereby improving the reliability of the regression results and providing robust evidence for the proposed hypotheses.

In the two-stage regression analysis using instrumental variables, the validity of the instruments was first tested. In the first stage, the coefficient of the instrumental variable for IPEn, measured by the number of patent applications, was 0.0080 and statistically significant at the 1 % level. This result indicates that the number of patent applications is an effective instrument for intellectual property empowerment. In the second stage, the regression analysis showed that the coefficient of IPEn on NQPF was 4.5536, significant at the 1 % level, suggesting a strong positive impact of IPEn on NQPF. Similarly, the coefficient of the digital DigEcon was 1.1388, also significant at the 1 % level, confirming that advancements in the digital economy significantly promote NQPF. The validity of the instrumental variables was further confirmed through statistical tests. The first-stage F-statistic was 9.93 (p-value = 0.0018), and both the Anderson Canonical Correlation LM statistic and the Cragg-Donald Wald F-statistic exceeded the critical threshold of 10, indicating the strength and validity of the instrumental variables. These results provide robust support for the causal relationships between IPEn, DigEcon, and NQPF (Table 9).

Table 8

PSM test.

VARIABLES	(1)	(2)
	NQPF	NQPF
IPEn	0.1793*** (4.1870)	
DigEcon		0.1331** (2.4781)
HCL	−2.2402 (−1.3210)	1.2485 (1.0713)
Open	−0.1557*** (−3.6795)	0.1423*** (4.1165)
Indus	−0.0955 (−0.9791)	−0.0785 (−1.1504)
Urban	0.3804* (1.9211)	0.6134*** (4.6624)
LnGDP	0.1501*** (3.3938)	0.1101*** (4.3730)
Constant	−1.4880*** (−3.5727)	−1.3566*** (−6.2799)
Observations	329	93
R-squared	0.286	0.823
Number of code	31	23
Pro	YES	YES
Year	YES	YES

t-statistics in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

Table 9

IV-2SLS.

VARIABLES	(1)	(2)	(3)	(4)
	first	second	first	second
	IPEn	NQPF		
IV	0.0080*** (3.1505)			
IPEn		4.5536*** (3.6170)		
IV1			−0.0000*** (−3.7783)	
DigEcon				1.1388*** (3.4153)
HCL	−3.1963*** (−3.6584)	4.8138 (1.1712)	−2.7819*** (−3.6692)	−2.1968* (−1.8653)
Open	0.0660** (2.2554)	0.1501 (0.9023)	0.2126*** (7.9866)	0.3476*** (4.0991)
Indus	0.1414*** (2.7979)	−0.1709 (−0.5306)	0.0813 (1.6497)	0.5801*** (7.5357)
Urban	0.1309 (1.6016)	−1.1830*** (−3.7245)	−0.3663*** (−4.5749)	−0.4951*** (−2.7654)
LnGDP	−0.0522*** (−2.7186)	0.4004*** (4.4465)	0.1459*** (7.2737)	0.0382 (0.6171)
Constant	0.4561** (2.5527)	−3.5742*** (−4.3969)	−1.2760*** (−6.7060)	−0.3521 (−0.6199)
First-Stage F-Statistic	9.93		14.28	
P value	0.0018		0.0002	
Anderson canon. corr. LM statistic		10.147		14.394
Cragg-Donald Wald F statistic		9.926		14.275
Observations	325	325	330	330
Pro	YES	YES	YES	YES
Year	YES	YES	YES	YES

t-statistics in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

4.6. Mechanism Verification

4.6.1. Mechanism testing based on intellectual property protection (IPP)

According to [hypotheses 2a and 2b](#), it is proposed that IPEn and the DigEcon enhance NQPF by strengthening IPP. The regression results support this proposition. The coefficient of IPEn on IPP is 0.1762 and statistically significant at the 1 % level, indicating that IPEn significantly enhances intellectual property protection. Similarly, the coefficient of DE on IPP is 0.4788, also significant at the 1 % level, suggesting that the digital economy substantially improves IPP by facilitating information flow and optimizing resource allocation. From a

theoretical perspective, enhanced intellectual property protection reduces technological infringement and increases the security of innovation outcomes, thereby incentivizing greater investment in R&D, ultimately boosting productivity. Furthermore, robust IPP attracts more investment and talent, fostering broader innovation activities that further contribute to the growth of new quality productivity. Thus, intellectual property protection plays a critical mediating role in the pathway through which the digital economy and intellectual property empowerment impact new quality productivity forces, underscoring its importance in this causal mechanism (Table 10).

4.6.2. Mechanism testing based on technological innovation (TI)

According to hypotheses 3a and 3b, IPEn and DigEcon promote NQPF through TI. At this stage, regression analysis was conducted to examine the effects of IPEn and DigEcon on technological innovation. The results reveal that IPE has a significant positive impact on TI, with a coefficient of 14.5760, significant at the 1 % level, indicating that IPE significantly fosters TI. Similarly, DigEcon also exhibits a significant positive effect on TI, with a coefficient of 10.2637, significant at the 1 % level, demonstrating that advancements in the digital economy substantially enhance technological innovation. Thus, technological innovation serves as a vital mediating variable, linking intellectual property empowerment, the digital economy, and new quality productivity forces, highlighting its pivotal role in this causal mechanism (Table 11).

Table 10
Mechanism testing for IPP.

VARIABLES	(1)	(2)
	IPP	IPP
IPEn	0.1762*** (2.8422)	
DigEcon		0.4788*** (5.5664)
HCL	−1.9897 (−0.8451)	1.2271 (0.5233)
Open	0.1367** (2.2912)	0.1800*** (3.1575)
Indus	−1.1458*** (−8.1616)	−0.9503*** (−6.7197)
Urban	−1.1764*** (−4.7055)	−0.9330*** (−3.8502)
LnGDP	0.1829*** (2.8965)	0.1081* (1.7494)
Constant	−0.2190 (−0.3608)	0.2568 (0.4379)
Observations	341	341
R-squared	0.739	0.758
Number of code	31	31
Pro	YES	YES
Year	YES	YES

t-statistics in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

Table 11
Mechanism testing for TI.

VARIABLES	(1)	(2)
	TI	TI
IPEn	14.5760*** (5.5887)	
DigEcon		10.2637*** (2.6300)
HCL	−149.6731 (−1.5108)	−84.4060 (−0.7933)
Open	−6.2970** (−2.5075)	−8.9813*** (−3.4731)
Indus	−0.8995 (−0.1523)	−0.1774 (−0.0276)
Urban	29.9247*** (2.8446)	38.9253*** (3.5402)
LnGDP	8.1126*** (3.0539)	5.7900** (2.0654)
Constant	−67.9333*** (−2.6599)	−49.3755* (−1.8553)
Observations	341	341
R-squared	0.316	0.261
Number of code	31	31
Pro	YES	YES
Year	YES	YES

t-statistics in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

4.7. Heterogeneity test of resource endowment

To explore the heterogeneity of resource endowments in the impact of the DigEcon and IPEn on NQPF, this section divides the sample into three categories: resource-based regions, agriculture-based regions, and industrial and service-based regions. The classification is as follows: (1) Resource-Based Regions: These regions rely primarily on the extraction and utilization of natural resources for economic development. They include provinces such as Shanxi, Inner Mongolia, Ningxia, Xinjiang, Qinghai, Shaanxi, Gansu, and Heilongjiang. Given their abundant natural resources, economic activities in these areas are closely tied to resource exploitation, categorizing them as resource-based regions. (2) Agriculture-Based Regions: The economies of these regions are predominantly driven by agriculture, focusing on the production and initial processing of agricultural products. This category includes provinces such as Henan, Hebei, Anhui, Jiangxi, Hunan, Guangxi, Yunnan, and Guizhou. These regions depend largely on agricultural activities for their economic foundation and are classified as agriculture-based regions. (3) Industrial and Service-Based Regions: These regions rely heavily on industrial and service sectors as their primary economic drivers. Provinces such as Beijing, Tianjin, Shanghai, Chongqing, Guangdong, Zhejiang, Jiangsu, Shandong, Fujian, Hubei, Sichuan, Hainan, Jilin, and Liaoning are included in this category. With strong industrialization and modern service industries, these regions exhibit advanced innovation capabilities and technological progress, making them industrial and service-based regions (Gao et al., 2024).

In the regression results, the coefficient of IPEn in resource-based areas is 0.3665, which is not significant at the 1 % significance level, indicating that in resource-based areas, the impact of IPEn on new productivity is weak. In agricultural areas, the coefficient of IPEn is 0.1404, and is significant at the 1 % significance level, indicating that intellectual property empowerment has a positive impact on the improvement of NQPF in agricultural areas. In industrial and service-oriented areas, the impact of IPEn on new productivity is more significant, with a coefficient of 0.2186, which is significant at the 1 % significance level, indicating that in these areas, IPEn has a significant impact on NQPF. The promotion effect is more obvious. DigEcon, the coefficient in resource-based areas is -0.5888 , and is not significant at the 1 % significance level, indicating that the impact of DigEcon on NQPF in resource-based areas is negative and not significant. In agricultural areas, the coefficient of digital economy is 0.2062, which is significant at the 1 % significance level, indicating that digital economy has a positive promotion effect on NQPF in agricultural areas. In industrial and service-oriented areas, the coefficient of the digital economy is 0.3127 and is significant at the 1 % significance level, indicating that the digital economy has a strong positive impact on NQPF in these areas (Table 12).

Table 12
Heterogeneity Test results.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	Resource	Agriculture	Industrial and Service	Resource	Agriculture	Industrial and Service
	NQPF	NQPF	NQPF	NQPF	NQPF	NQPF
IPEn	0.3665 (0.8165)	0.1404 (1.3507)	0.2186*** (6.1499)			
DigEcon				-0.5888 (-1.4244)	0.2062*** (4.1931)	0.3127*** (4.8172)
HCL	-6.2478 (-0.9496)	-3.4333^{**} (-2.3088)	-1.4990 (-0.7142)	-7.2422 (-1.1089)	-1.7366 (-1.2405)	2.1346 (0.8749)
Open	-0.4468^{*} (-1.8748)	0.0873 (0.9019)	-0.0799^{*} (-1.9182)	-0.3945 (-1.6497)	0.1001 (1.1619)	-0.1013^{**} (-2.3146)
Indus	0.3695 (1.0241)	0.1339 (1.0627)	0.0297 (0.1874)	0.1346 (0.4265)	0.1401 (1.2461)	0.0144 (0.0862)
Urban	1.5326*** (2.0814)	-0.3519 (-1.6176)	0.1699 (0.8214)	1.7031** (2.2930)	-0.3446^{*} (-1.7643)	0.3825* (1.8255)
LnGDP	-0.2202 (-0.9073)	0.0697** (2.2011)	0.2450*** (4.3626)	-0.0688 (-0.3374)	0.0531* (1.8577)	0.1667*** (2.9027)
Constant	1.6069 (0.6895)	-0.4656 (-1.5540)	-2.4360^{***} (-4.1953)	0.0928 (0.0468)	-0.3407 (-1.2689)	-1.8128^{***} (-3.0418)
Observations	88	88	154	88	88	154
R-squared	0.163	0.698	0.643	0.180	0.756	0.608
Number of code	8	8	14	8	8	14
Pro	YES	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES	YES

5. Conclusion

This study provides an in-depth analysis of the dual roles of intellectual property empowerment (IPEn) and the digital economy (DigEcon) in advancing new quality productivity forces (NQPF), emphasizing their multidimensional impacts, underlying mechanisms, regional heterogeneity, and threshold effects, thus contributing to the growing body of knowledge on innovation-driven economic development. Both IPEn and DigEcon demonstrate significant positive effects on NQPF, as validated by robust empirical evidence, underscoring their independent and complementary roles in fostering high-quality productivity forces. Mechanism testing highlights the importance of intellectual property protection (IPP) and technological innovation (TI) as critical mediators, where enhanced IPP reduces innovation risks, safeguards intellectual property rights, and facilitates R&D investment, while TI drives technological progress, industrial upgrades, and the creation of new products and services (Cao et al., 2023; Rao & Liu, 2024).

The findings also reveal marked regional heterogeneity, with IPEn and DigEcon exerting the most pronounced effects in industrial and service-oriented regions, where higher levels of technological absorption, infrastructure, and institutional support enhance their productivity-enhancing potential. In contrast, their impacts are weaker or negative in resource-based regions, reflecting the constraints imposed by resource dependence, limited diversification, and structural inefficiencies. This underscores the importance of tailoring policies to regional characteristics to fully unlock the potential of IPEn and DigEcon. Furthermore, the results illustrate a nonlinear relationship between IPEn, DigEcon, and NQPF, mediated by industrial structure rationalization. In regions with well-rationalized structures, these factors yield substantial NQPF gains, while imbalanced structures suppress their effectiveness, highlighting the critical role of achieving balanced economic development.

From an academic perspective, this study makes several important contributions to literature. First, it enriches the theoretical framework of productivity growth by integrating intellectual property and digital economy perspectives, addressing gaps in understanding their joint and independent effects on new high-quality productivity forces. Second, it extends the methodological toolkit by employing threshold effect models to capture the nonlinear dynamics of industrial structure rationalization, offering a nuanced understanding of the conditions under which IPEn and DigEcon are most effective. Third, it provides empirical evidence for the mediating roles of IPP and TI, advancing knowledge on how these factors translate into NQPF gains. Finally, by identifying regional heterogeneity and resource endowment effects, the study contributes to a more granular understanding of the spatial dimensions of economic development.

6. Implications, limitations and future research directions

This study offers several important implications for policy and practice regarding the enhancement of new quality productive forces in China. The findings highlight the necessity of further strengthening intellectual property protection mechanisms and promoting the effective utilization of intellectual property assets. In particular, improving the legal framework for intellectual property protection and enhancing enforcement efficiency are crucial for reducing the costs of patent application, transaction, and enforcement, especially for small and medium-sized enterprises. Such improvements would not only encourage firms to increase their research and development (R&D) investment but also facilitate the transformation of innovative achievements into commercial value, thereby fostering a virtuous cycle of innovation and productivity growth.

In parallel, accelerating the development of digital infrastructure is essential to ensure the broad-based growth of the digital economy. Priority should be given to improving connectivity and technological penetration in underdeveloped regions, particularly resource-based areas in western China. Bridging the digital divide can enhance regional inclusiveness and unlock latent productivity potential. Furthermore, the deep integration of digital technologies—such as big data, cloud computing, and artificial intelligence—into traditional industries is vital for industrial upgrading and the realization of intelligent manufacturing, which collectively contribute to sustainable economic development.

The empirical results underscore notable regional heterogeneity, indicating that intellectual property empowerment and digital economy development have the most pronounced effects in industrial and service-oriented regions. Therefore, regional policies must be adapted to local conditions. Advanced regions should focus on leading innovation-driven development, while resource-dependent areas require tailored strategies to promote economic diversification, reduce reliance on natural resources, and facilitate a transition toward knowledge- and technology-based industries.

Moreover, fostering a synergistic innovation mechanism that combines intellectual property empowerment with digital economy initiatives is critical. Establishing an integrated system that facilitates the interaction between intellectual property protection, technological innovation, and data governance will support the formation of an efficient innovation ecosystem. For example, promoting data assetization and ensuring secure data transactions can complement intellectual property strategies, enabling the emergence of new forms of intellectual property rights such as data patents and algorithm copyrights. This interaction reinforces the co-evolution of data flows and knowledge production, optimizing resource allocation and enhancing the value of digital resources.

At the enterprise level, policies should aim to strengthen firms' roles as the primary agents of innovation by encouraging their participation in the commercialization of intellectual property and digital transformation. Governments can provide targeted support through fiscal incentives, funding programs, and tax benefits to lower barriers to innovation and facilitate knowledge diffusion. Additionally, fostering cooperation between enterprises, research institutions, and financial organizations will create a multi-layered support system that accelerates the commercialization of technological advances.

Lastly, constructing a sound institutional environment through policy coordination and legal reforms is essential to sustain the long-term development of new quality productive forces. Policymakers should ensure that intellectual property protection, digital economy regulations, and industrial policies are coherent and mutually reinforcing. This includes enhancing cross-regional and cross-

sectoral coordination, as well as participating in the establishment of international rules for intellectual property protection and digital economy governance, thereby strengthening China's position in the global innovation landscape.

Despite the valuable findings of this study, several limitations remain. First, there are data constraints. The dataset employed in this research may suffer from limited time spans or incomplete sample coverage, which imposes certain restrictions on the generalizability of the conclusions. Second, the regional classification adopted in this study—covering industrial and service-oriented regions, agricultural regions, and resource-dependent regions—remains relatively broad. It fails to further distinguish the specific characteristics of sub-regional types, which may obscure potential heterogeneity across different areas.

In light of the findings and limitations identified in this study, future research may advance in the following directions. First, data expansion is necessary. By collecting data with a longer time horizon and covering a broader range of regions, researchers can better validate the robustness and universality of the study's conclusions. Second, model optimization should be pursued. Future studies could incorporate additional influencing variables—such as the policy environment and the dormancy period of corporate intellectual property rights—to develop more sophisticated models that offer a more comprehensive depiction of the relationship between intellectual property (IP) empowerment, the digital economy, and the development of new quality productive forces. Third, regional differentiation should be further refined. By subdividing regional types—such as technology-driven cities and traditional industrial cities—future research could explore the heterogeneity in the synergy between intellectual property and the digital economy across various regions. Fourth, there is a need for deeper investigation into the underlying mechanisms. Specifically, future studies should provide a detailed analysis of how the combination of intellectual property and the digital economy drives the development of new quality productive forces, illustrating the “dual-engine” model of intellectual property and digitalization.

Future research should focus on deconstructing the multidimensional pathways through which this mechanism operates. On the one hand, it is essential to reveal how intellectual property transformation—through mechanisms such as technological certification, value circulation, and ecosystem reconstruction—provides institutional guarantees and innovative momentum for the digitalization process. On the other hand, attention should be paid to how the property rights and assetization of data elements feed back into the intellectual property system, giving rise to new forms of rights, such as “data patents” and “algorithm copyrights.” This interaction fosters a symbiotic logic in which the data flow nurtures knowledge production, while the knowledge flow optimizes data value. Future research could apply dynamic spatial panel models to investigate the spillover effects of new quality productivity across regions, and utilize firm-level microdata to enhance model precision. In addition, in-depth analysis of the interactive mechanisms between intellectual property and the digital economy is needed to explore their nonlinear effects across different stages of industrial transformation.

This study not only affirms the productivity-enhancing value of intellectual property transformation but also reveals the mechanism of new quality productive force transition and regional adaptation under the dual empowerment perspective. Against the backdrop of dual transformations in the digital and green economies, intellectual property governance has evolved beyond a legal proposition to become a crucial arena for national strategic capability competition. The “dual-engine” model of intellectual property and digitalization is reshaping the global innovation landscape. In the emerging paradigm of deep integration between the digital and knowledge economies, the dual-engine effect of intellectual property and digitalization is shifting from a simple linear addition to a complex nonlinear synergy. Future research should strive for continuous breakthroughs in frontier areas such as the construction of dynamic adaptive models and the design of composite enhancement frameworks for new quality productivity forces. Such endeavors aim to provide a Chinese solution that integrates theoretical rigor with practical relevance, contributing to the formation of a new development pattern.

Declaration of interest statement

The authors declare that no competing interests exist.

Ethic statement

Not applicable.

Author Statement

1. Yiling Liu: Conceptualization, Methodology, Data curation, Writing- Original draft preparation.
2. Yu-Cheng Lin: Visualization, Investigation.

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Data availability

All relevant data are available from the authors on request.

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